

Chapter 7: Model Checking in Parametric Models

Modelling Survival Data in Medical Research

Collett (2023), 4th ed.

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Residuals for parametric models

1. Standardised residuals
2. Cox–Snell residuals
3. Martingale & deviance residuals
4. Score residuals
5. Weibull, log-logistic, lognormal forms

Assessing model fit

6. Analysis of residuals
7. Observed vs. fitted survivor functions

Influence and assumptions

8. Influence on a single parameter
9. Influence on all parameters
10. Testing proportional hazards

Goal: assess whether a fitted parametric survival model adequately describes the data, and detect observations that distort it.

Why Model Checking?

Diagnostics matter just as much here:

Assessing model adequacy is as important for parametric models as for the Cox model. Many parametric residuals build on those developed for the Cox model.

The accelerated failure time setup:

For the i th individual,

$$\log T_i = \mu + \alpha_1 x_{1i} + \cdots + \alpha_p x_{pi} + \sigma \epsilon_i$$

where the distribution of ϵ_i depends on the distribution chosen for T_i .

What we will check:

- ▶ Whether the assumed **distribution** fits (residual plots)
- ▶ Whether the **fitted survivor function** matches the data
- ▶ Whether any **observations** unduly influence the estimates
- ▶ Whether the **proportional hazards** assumption holds

A complication: censoring

If an individual's survival time is censored, its residual is also censored — which complicates the interpretation of residuals.

Definition:

A natural “observation minus fitted value” residual:

$$r_{Si} = \frac{\log t_i - \hat{\mu} - \hat{\alpha}_1 x_{1i} - \cdots - \hat{\alpha}_p x_{pi}}{\hat{\sigma}}$$

where t_i is the observed time and $\hat{\mu}, \hat{\sigma}, \hat{\alpha}_j$ are the fitted parameters.

Expected behaviour:

If the model is correct, the r_{Si} behave like a (possibly censored) sample from the distribution of ϵ_i .

Example: the Weibull model

If T_i is Weibull, then ϵ_i has a **Gumbel** distribution, so the standardised residuals should look like a Gumbel sample.

The key diagnostic link:

By a general result, $-\log S_{\epsilon_i}(\epsilon)$ has a **unit exponential** distribution. So if the model fits,

$$-\log S_{\epsilon_i}(r_{Si}) \approx \text{unit exponential}$$

This is the basis for the Cox–Snell diagnostic plot used to assess overall model adequacy.

Definition:

The estimated cumulative hazard at the observed time — exactly as for the Cox model, but now using **parametric** functions:

$$r_{Ci} = \hat{H}_i(t_i) = -\log \hat{S}_i(t_i)$$

where the estimated survivor function (from the AFT model) is

$$\hat{S}_i(t) = S_{\epsilon_i} \left(\frac{\log t - \hat{\mu} - \hat{\alpha}_1 x_{1i} - \cdots - \hat{\alpha}_p x_{pi}}{\hat{\sigma}} \right)$$

Key property:

If the correct model is fitted, the Cox–Snell residuals have a **unit exponential** distribution (censored observations give censored residuals).

Relationship to standardised residuals:

$$r_{Ci} = -\log S_{\epsilon_i}(r_{Si})$$

Checking whether the standardised residuals follow the right distribution is **equivalent** to checking whether the Cox–Snell residuals are unit exponential.

Definition:

The difference between the observed event indicator and the Cox–Snell residual:

$$r_{Mi} = \delta_i - r_{Ci}$$

where $\delta_i = 1$ for an event, 0 for a censored observation.

Interpretation:

A measure of the difference between the **observed** number of events in $(0, t_i)$ — either 0 or 1 — and the number the model **predicts**.

Spotting poor fit:

Observations with unusually **large** martingale residuals are poorly fitted by the model.

Properties:

- ▶ They **sum to zero**
- ▶ But are **not symmetric** about zero (skewed)
- ▶ Range from $-\infty$ to a maximum of 1

A note on the name

Strictly, the martingale derivation does not carry over to the AFT model, but the term is kept for convenience.

Definition:

A transformation of the martingale residuals to make them more **symmetric** about zero:

$$r_{Di} = \text{sgn}(r_{Mi}) [-2\{\delta_i \log(\delta_i - r_{Mi})\}]^{1/2}$$

These are often similar to the deviance residuals from a comparable Cox model.

A sign convention for AFT models:

Because the AFT model is built on **log survival time** rather than the hazard, r_{Di} has the *opposite* sign from the standardised residual r_{Si} . So for AFT models, the deviance residual is more appropriately defined as $-r_{Di}$ (as some software does).

A note on the name

These are *not* components of the deviance for the fitted model, but the name is retained for convenience.

Use deviance residuals to identify poorly-fitted observations in index plots and plots against time.

Definition:

The components of the derivatives of the log-likelihood with respect to each parameter, evaluated at the MLEs.

With $z_i = (\log t_i - \mu - \alpha'x_i)/\sigma$, the derivatives are

$$\frac{\partial \log L}{\partial \mu} = \sigma^{-1} \sum_i g(z_i)$$

$$\frac{\partial \log L}{\partial \sigma} = \sigma^{-1} \sum_i \{z_i g(z_i) - \delta_i\}$$

$$\frac{\partial \log L}{\partial \alpha_j} = \sigma^{-1} \sum_i x_{ji} g(z_i)$$

where

$$g(z_i) = (1 - \delta_i) \frac{f_{\epsilon_i}(z_i)}{S_{\epsilon_i}(z_i)} - \delta_i \frac{f'_{\epsilon_i}(z_i)}{f_{\epsilon_i}(z_i)}$$

The score residual for covariate X_j :

$$r_{Uji} = \hat{\sigma}^{-1} x_{ji} g(r_{Si})$$

Key facts:

- ▶ The score residuals for the covariates are the most important
- ▶ Because they are the derivatives at the maximum, they **sum to zero**
- ▶ They parallel the Schoenfeld residuals of the Cox model

Residuals: The Weibull Distribution

The error distribution:

For the Weibull AFT model, ϵ_i has a **Gumbel** distribution with survivor function

$$S_{\epsilon_i}(\epsilon) = \exp(-e^\epsilon)$$

So the standardised residuals should behave like a (possibly censored) Gumbel sample.

Cox–Snell residuals (very simple here):

$$r_{Ci} = -\log S_{\epsilon_i}(r_{Si}) = \exp(r_{Si}),$$

just the exponentiated standardised residuals.

Score residual for covariate X_j :

After substituting the Gumbel density (where $g(z_i) = e^{z_i} - \delta_i$):

$$r_{Uji} = \hat{\sigma}^{-1} x_{ji} (e^{r_{Si}} - \delta_i)$$

A useful consequence:

The score residual for μ equals $\hat{\sigma}^{-1}(r_{Ci} - \delta_i)$. Since the score residuals sum to zero, the **martingale residuals must also sum to zero** in the Weibull model.

The martingale and deviance residuals follow directly from the Cox–Snell residuals via the standard formulas.

Residuals: Log-logistic and Lognormal

Log-logistic model:

ϵ_i has a **logistic** distribution,
 $S_{\epsilon_i}(\epsilon) = (1 + e^\epsilon)^{-1}$. So:

$$r_{Ci} = \log\{1 + \exp(r_{Si})\}$$

$$r_{Uji} = \hat{\sigma}^{-1} x_{ji} \frac{\exp(r_{Si}) - \delta_i}{1 + \exp(r_{Si})}$$

The standardised residuals should look like a logistic sample if the model fits.

Lognormal model:

ϵ_i has a **normal** distribution, so
 $S_{\epsilon_i}(\epsilon) = 1 - \Phi(\epsilon)$:

$$r_{Ci} = -\log\{1 - \Phi(r_{Si})\}$$

$$r_{Uji} = \hat{\sigma}^{-1} \left[\frac{(1 - \delta_i) f_{\epsilon_i}(r_{Si})}{1 - \Phi(r_{Si})} + \delta_i r_{Si} \right]$$

where f_{ϵ_i} is the standard normal density and Φ its CDF.

In every case the recipe is the same: the error distribution S_{ϵ_i} determines the Cox–Snell and score residuals; the rest follow from the general formulas.

The primary diagnostic plot:

Compare the distribution of the **Cox–Snell residuals** with the unit exponential:

- 1 Compute the Kaplan–Meier estimate $\hat{S}(r_{Ci})$ of the residuals' survivor function
- 2 Plot $-\log \hat{S}(r_{Ci})$ against r_{Ci}

A **straight line through the origin with unit slope** suggests the model fits. (Or plot $\log\{-\log \hat{S}(r_{Ci})\}$ against $\log r_{Ci}$ — also a unit-slope line through the origin.)

Why this plot is more trustworthy here:

The criticisms levelled at this plot for the Cox model have **less force** for parametric models: the non-parametric baseline is replaced by a parametric function with just a few parameters (μ, σ) , so the residuals are more likely to be genuinely unit exponential.

Other useful plots:

- ▶ Index plots of **martingale/deviance** residuals — spot poorly-fitted observations
- ▶ These residuals vs. time, rank, or covariates — spot where the model fails
- ▶ Residuals vs. the risk score / acceleration factor
- ▶ **Score residual** plots for a fuller assessment

Example: Ovarian Cancer — Cox–Snell Plot

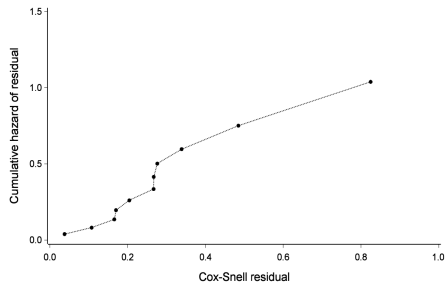


Figure: Cumulative hazard plot of Cox–Snell residuals from a Weibull model fitted to the ovarian cancer data.

The fitted model:

A Weibull PH model with *Age* and *Treat*. In AFT form, $\hat{S}_i(t) = \exp\{-\exp(z_i)\}$ where

$$z_i = \frac{\log t - 10.4254 + 0.0790 \text{ Age}_i - 0.5615 \text{ Treat}_i}{0.5489}$$

with Cox–Snell residuals $r_{Ci} = \exp(r_{Si})$.

Reading the plot:

- ▶ The points lie close to a line of intercept ≈ 0 and slope ≈ 1 — broadly supportive
- ▶ But there is some **systematic deviation** from the line, giving mild cause for concern about the fit

Example: Ovarian Cancer — Martingale Residuals

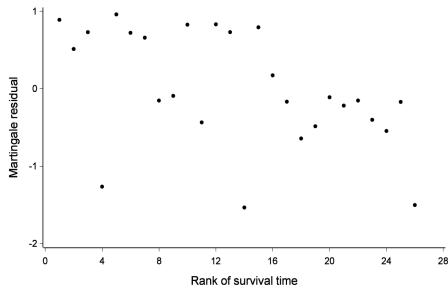


Figure: Martingale residuals plotted against the rank order of survival times.

Interpretation:

- ▶ Residuals are asymmetric and bounded above by 1.
- ▶ Useful for detecting outliers and assessing functional form.
- ▶ A slight tendency for longer survival times to have smaller residuals.
- ▶ Most long survivors are censored observations.

No major departure from model assumptions is evident, although some observations are less well fitted.

Example: Ovarian Cancer — Deviance Residuals

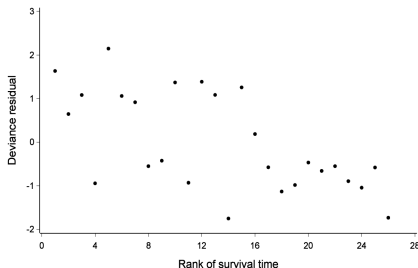


Figure: Deviance residuals plotted against the rank order of survival times.

Interpretation:

- ▶ Deviance residuals transform martingale residuals into a more symmetric scale.
- ▶ Easier to identify observations that are poorly fitted.
- ▶ A few observations exhibit relatively large residuals.
- ▶ No strong systematic trend is apparent.

Deviance residuals suggest a generally adequate fit, with a small number of potentially influential observations.

Example: Ovarian Cancer — Score Residuals

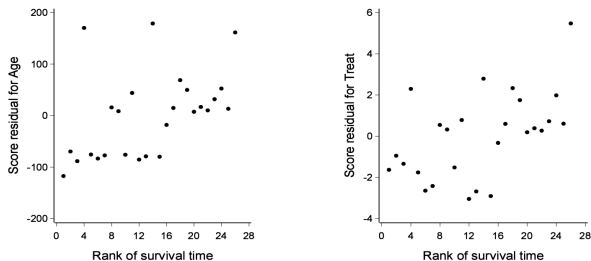


Figure: Score residuals for Age and Treat plotted against the rank order of survival times.

Interpretation:

- ▶ Score residuals measure each observation's contribution to coefficient estimation.
- ▶ For *Age*, patients 14, 4, and 26 have relatively large residuals.
- ▶ For *Treat*, patient 26 is particularly noticeable.
- ▶ These observations may have greater influence on the fitted model.

Patients 14, 4, and 26 appear less well fitted and merit further investigation.

Comparing Observed and Fitted Survivor Functions

A direct visual check:

In parametric modelling, the fitted survivor function is a **smooth, continuous** function of t — so it can be plotted and compared with the observed (Kaplan–Meier) estimate.

Simple cases:

- ▶ **Single sample, no covariates:** compare the fitted curve directly with the Kaplan–Meier step function
- ▶ **One or two factors:** for each group (e.g. treatment $\times$ gender), compare the fitted curve with that group's Kaplan–Meier estimate

Many explanatory variables:

With several covariates, we cannot form a group per covariate pattern. Instead, group individuals by their **risk score**:

- ▶ PH model: the risk score is $\hat{\beta}'x_i$
- ▶ AFT model: use $\hat{\alpha}'x_i$ or the acceleration factor

Large positive risk scores correspond to greater hazard. Grouping by risk score lets us compare observed and fitted survival even with many covariates.

The procedure:

- 1 Sort individuals by their risk score
- 2 Divide into groups (3 for small data; 5 or 7 for larger) — e.g. 5 groups give 20% each, from low to high risk
- 3 For group j , average the fitted survivor functions:

$$\bar{S}_j(t) = \frac{1}{n_j} \sum_{i=1}^{n_j} \hat{S}_{ij}(t)$$

- 4 Overlay the group's **Kaplan–Meier** estimate

Why it is useful:

Departures from the model are often **easier to detect** this way than from residual plots.

Limitations

- ▶ The **same model** defines the groups *and* the fitted curves — ideally split the data (fit on one half, check on the other)
- ▶ Individuals with different covariates but the same risk score are treated alike

Example: Ovarian Cancer — Observed vs. Fitted

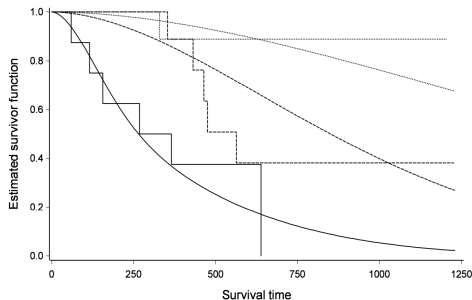


Figure: Observed (Kaplan–Meier) and fitted survivor functions for low- (.....), medium- (- - -), and high-risk (—) ovarian cancer patients. The observed survivor function is the step function.

The risk score:

$\hat{\eta}_i = 0.144 \text{ Age}_i - 1.023 \text{ Treat}_i$, used to split the 26 patients into three groups.

Findings:

- ▶ **High-risk group:** good agreement; the model fits well
- ▶ **Middle group:** fitted survival runs a little *too high*; these patients also have the largest martingale residuals
- ▶ **Low-risk group:** only one death, so little can be concluded

The grouped comparison localises the lack of fit to the medium-risk patients; consistent with the residual analysis.

Identifying Influential Observations

Two kinds of influence to check:

- 1 Influence on a **single** parameter estimate
- 2 Influence on the **whole set** of parameter estimates

The ingredients:

Diagnostics are computed from all $p + 2$ parameter estimates and their variance–covariance matrix.
Collect the parameters into

$$\theta' = (\mu, \alpha_1, \dots, \alpha_p, \sigma)$$

Derived from the AFT representation:

The diagnostics are developed for the Weibull PH model via its accelerated failure time form, but adapt to other models in the family.

The same logic as influence diagnostics for the Cox model: ask how much each observation moves the estimates, using the score residuals and the variance–covariance matrix.

The delta-beta statistic:

An approximation to the change in $\hat{\theta}_j$ when the i th observation is omitted:

$$\Delta_i \hat{\theta}_j = \text{the } j\text{th component of } \text{var}(\hat{\theta}) u_i$$

where:

- ▶ $\text{var}(\hat{\theta})$ is the estimated variance–covariance matrix
- ▶ u_i is the $(p + 2) \times 1$ vector of **score residuals** for observation i

Delta-betas for the covariate coefficients:

The components for the α_j are called **delta-betas**. Dividing by the standard error of $\hat{\alpha}_j$ gives **standardised** delta-betas.

How to use them:

- ▶ Index plots, or plots against rank order of survival time
- ▶ Large values flag observations that strongly affect that particular coefficient

A delta-beta answers: “how much would this coefficient change if I dropped this one patient?”

Influence on All Parameters

Likelihood displacement:

The difference in $-2 \log \hat{L}$ between the model fitted to the full data and to the data with observation i removed. Obtained from the diagonal of $U \text{var}(\hat{\theta}) U'$, where U holds the score residuals. Large values flag observations that influence the overall fit.

The F_i statistic (Hall, Rogers & Pregibon):

$$F_i = \frac{u_i' R^{-1} u_i}{(p+2)\{1 - u_i' R^{-1} u_i\}}$$

where $R = \sum_i u_i u_i'$ is the cross-product matrix of score residuals.

The C_i statistic:

$$C_i = \frac{u_i' \text{var}(\hat{\theta}) u_i}{\{1 - u_i' \text{var}(\hat{\theta}) u_i\}^2}$$

Using F_i and C_i :

- ▶ Their numerical values typically differ from each other
- ▶ In each case, a **relatively large** value flags an influential observation
- ▶ To see *how* it influences the fit, omit it and refit the model

Likelihood displacement, F_i , and C_i each summarise how much one observation moves the **entire** set of estimates.

Example: Ovarian Cancer — Delta-Betas for Age

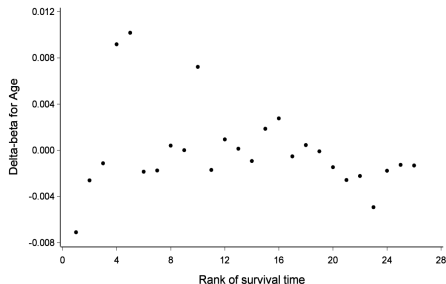


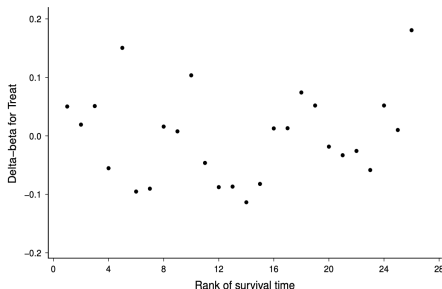
Figure: Delta-betas for Age plotted against the rank order of survival times.

Interpretation:

- ▶ Two observations stand out: patients 4 and 5.
- ▶ Both have unusually short survival times.
- ▶ One patient is relatively old (74 years), the other relatively young (43 years).

Patients 4 and 5 exert the strongest influence on the estimated **Age** coefficient.

Example: Ovarian Cancer — Delta-Betas for Treat



Interpretation:

- ▶ No observations stand out as highly influential.
- ▶ Changes in the treatment coefficient are small when individual patients are removed.
- ▶ The treatment effect appears relatively stable.

No single patient has a substantial influence on the estimated **Treat** coefficient.

Figure: Delta-betas for Treat plotted against the rank order of survival times.

Example: Ovarian Cancer — F Influence Statistic

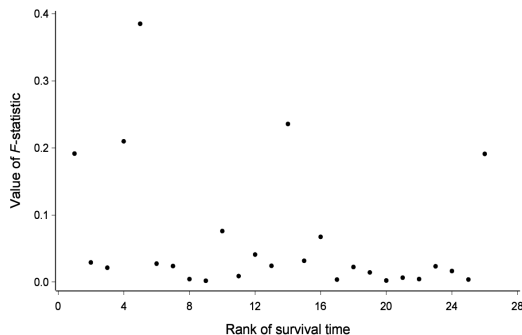


Figure: F influence statistic plotted against the rank order of survival times.

Key finding:

- ▶ Patient 5 is clearly influential
- ▶ Patients 1, 4, 14, and 26 also merit attention

Example: Ovarian Cancer — C Influence Statistic

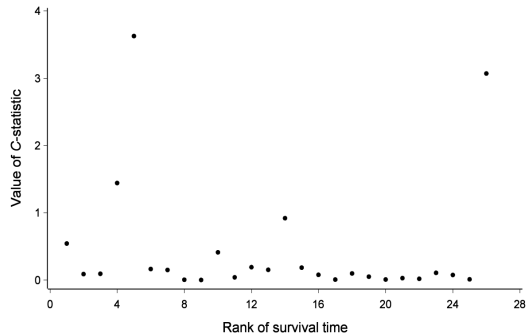


Figure: C influence statistic plotted against the rank order of survival times.

Key finding:

- ▶ Patients 5 and 26 stand out as most influential

Testing Proportional Hazards in the Weibull Model

The key insight:

Across g groups, proportional hazards holds if and only if the Weibull **shape parameter** γ is **the same** in every group.

The likelihood ratio test:

- 1 Fit a **separate** Weibull model to each group (same linear predictor, but its own γ and λ). Sum the $-2 \log \hat{L}$ values $\rightarrow -2 \log \hat{L}_1$
- 2 Fit **one** Weibull PH model to the combined data with group effects and interactions (a **common** γ) $\rightarrow -2 \log \hat{L}_0$

The test statistic:

$$-2 \log \hat{L}_0 - (-2 \log \hat{L}_1) \sim \chi^2_{g-1}$$

This is the change in $-2 \log \hat{L}$ from constraining the shape parameters to be equal.

Decision:

If the difference is **not significant**, the common-shape (proportional hazards) assumption is justified.

Note: parametric models with time-dependent covariates could also test PH, but software for them is not widely available.

Example: Ovarian Cancer — Testing PH

The two treatment groups:

Fit a Weibull model containing *Age* separately to each treatment group:

Group	$-2 \log \hat{L}$
Single treatment	22.851
Combined treatment	16.757
Sum ($-2 \log \hat{L}_1$)	39.608

The common-shape model:

A model with separate age effects per group, a treatment effect, and a **common** shape parameter gives

$$-2 \log \hat{L}_0 = 39.708$$

The test:

$$39.708 - 39.608 = 0.10 \sim \chi_1^2$$

This is **not significant**.

The two shape parameters may be taken as equal — the proportional hazards assumption is justified for these data.

Chapter Summary

- ▶ Model checking is as important for **parametric** survival models as for the Cox model. Most diagnostics are based on the AFT form $\log T_i = \mu + \alpha'x_i + \sigma\epsilon_i$.
- ▶ **Standardised residuals** $r_{Si} = (\log t_i - \hat{\mu} - \hat{\alpha}'x_i)/\hat{\sigma}$ should follow the assumed error distribution (Gumbel for Weibull, logistic for log-logistic, normal for lognormal).
- ▶ **Cox–Snell residuals** $r_{Ci} = -\log \hat{S}_i(t_i)$ should be unit exponential under a correct model. This is checked with a cumulative hazard plot through the origin with slope one.
- ▶ **Martingale** $(\delta_i - r_{Ci})$ and **deviance** residuals detect poor fit. **Score residuals** relate to Schoenfeld residuals and support influence analysis.
- ▶ Comparing **Kaplan–Meier** curves with fitted survivor functions by risk group often reveals misfit more clearly than residual plots.
- ▶ **Influence diagnostics:** delta-betas measure the effect on a single coefficient; likelihood displacement, F_i , and C_i measure impact on all parameters.
- ▶ **Weibull PH check:** proportional hazards holds if the shape parameter γ is common across groups, tested via a χ^2_{g-1} likelihood ratio test.

Questions?

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